

Project Report

1. INTRODUCTION

1.1 Project Overview

The Heart Disease Analysis project is an end-to-end data processing and business intelligence solution designed to analyze a clinical dataset comprising 4,416 patient records. By migrating raw data into a relational database engine and linking it to a modern data visualization platform, this project uncovers hidden correlations between behavioral, physiological, and demographic factors and the onset of heart disease.

1.2 Purpose

The secondary goal of this project is to eliminate technical barriers in health informatics, allowing non-technical medical professionals, policy planners, and patients to extract instant health risk insights. By replacing manual, row-by-row datasheet reviews with interactive dashboards, it speeds up clinical risk profiling and supports data-driven public health decisions

2. IDEATION PHASE

2.1 Problem Statement

Raw, unstructured, or tabular clinical healthcare records are difficult for medical practitioners and health policy makers to interpret quickly. Without clear visual systems, spotting critical high-risk patient combinations such as specific age groups combined with high BMI and lifestyle indicators like smoking is slow and error-prone, delaying preventative medical care and complicating public health resource planning

2.2 Empathy Map Canvas

Our design process centered on understanding three key primary stakeholder personas:

- **Dr. Sharma (Clinician):** Says he needs to identify high-risk patients quickly; **Thinks** manual table reading wastes time; **Does** cross-reference lifestyle data against cardiac histories; **Feels** frustrated by complex database software interfaces.
- **Ramesh (Public Health Official):** Says data must back up regional policy decisions; **Thinks** about geographic discrepancies; **Does** request regional demographic splits; **Feels** pressured to optimize limited budgets.
- **Anita (Patient):** Says she wants to track her personal medical risks; **Thinks** medical terminology is confusing; **Does** look up health standards; **Feels** anxious about complex diagnostic layouts.

2.3 Brainstorming

During the ideation phase, the team focused on addressing database connectivity, interface simplicity, and analytical depth. We decided to move away from text reports in favor of a 1-click cross-filtering dashboard UI. Key elements identified during brainstorming included converting technical database shortcodes into natural language labels and designing custom calculated filters to isolate patient lifestyle risk brackets automatically.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

The target analytical journey moves from raw data extraction to actionable healthcare insight:

Data Intake: Data analysts import the cohort data into a structured system, verifying table counts.

Interface Interaction: Clinicians and policy makers log in to access the analytical visual layers.

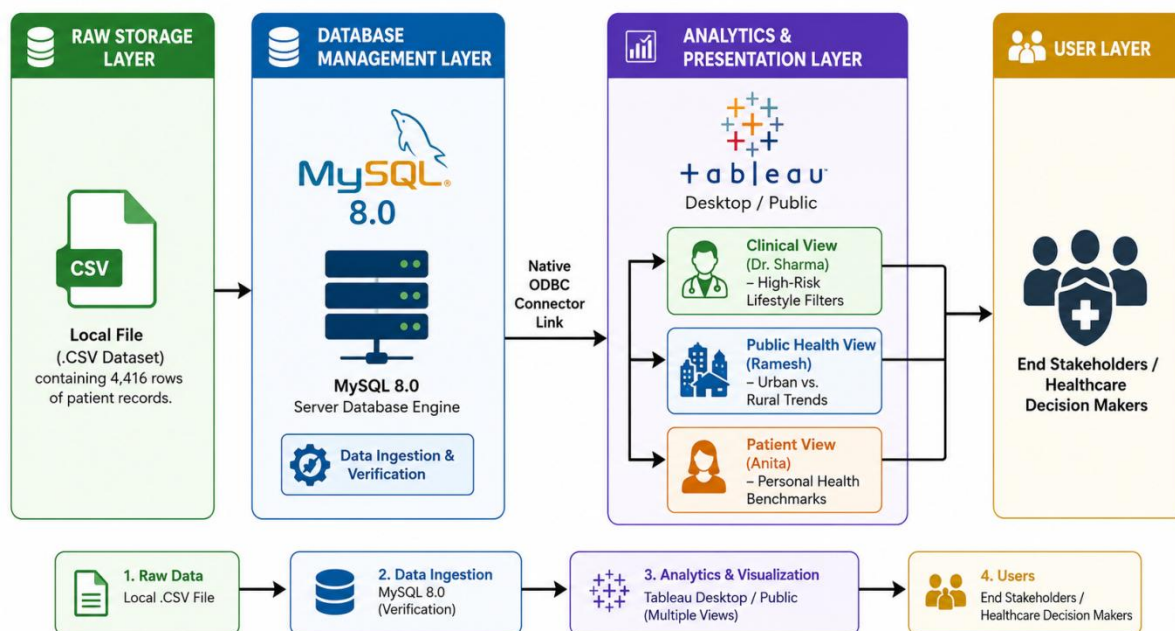
Cross-Filtering: Users apply interactive multi-perspective filters to narrow data down by age, region, or habit.

Clinical/Policy Decision: Stakeholders export or view the resulting trends to guide patient care or regional budgets.

3.2 Solution Requirement

- **Functional Requirements:** The system must import and store 4,416 historical patient rows; dynamically calculate BMI categories; filter health data by age group and smoking status; and display regional health trends.
- **Non-Functional Requirements:** The dashboard must load filter views in under 2 seconds; maintain data integrity across database-to-visual interfaces; and use an intuitive, user-friendly layout.

3.3 DataFlow Diagram:



3.4 Technology Stack

- Database Engine: MySQL 8.0 Server (for relational storage, verification, and schema integrity).
- Data Connectivity Interface: Tableau Native ODBC Database Driver.
- Analytics & BI Presentation Layer: Tableau Desktop / Tableau Public (for visual dashboard builds and storyboards)

4. PROJECT DESIGN

4.1 Problem Solution Fit

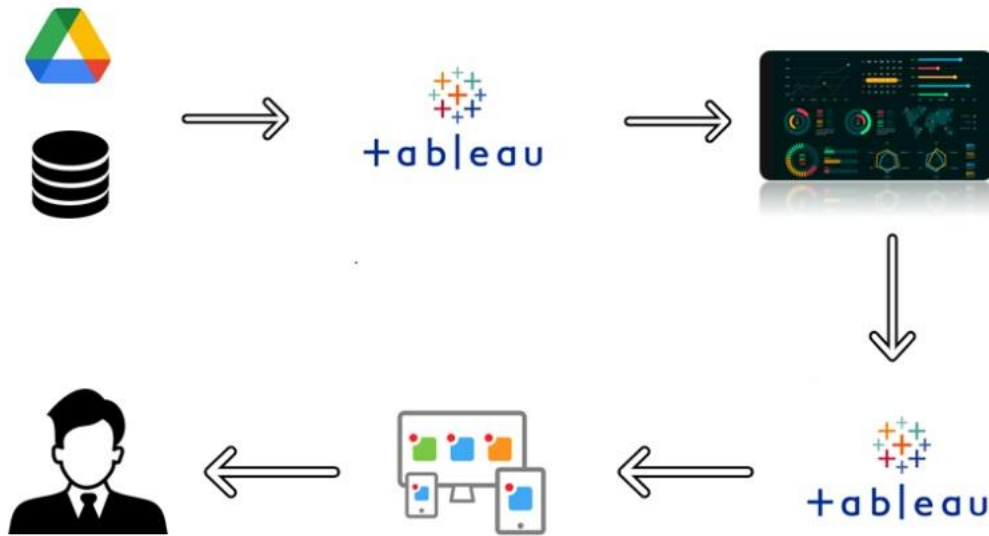
The proposed solution directly addresses our problem statement by mapping raw, complex columns to clean visual chart properties. Instead of querying rows manually, Dr. Sharma can use 1-click dashboard filters to identify high-risk groups, and Ramesh can access instant demographic breakdowns to support health policy planning.

4.2 Proposed Solution

An integrated database-to-dashboard visualization pipeline that processes 4,416 patient records and presents

them across three functional, user-specific views. The system renames technical columns to clean language and uses custom calculated fields to categorize clinical variables like BMI and age ranges.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

Project Planning

The project was executed in structured sprint intervals over a fast-paced timeline running from May 25, 2026, to July 07, 2026, with project delivery kicking off on July 03, 2026

6. FUNCTIONAL AND PERFORMANCE TESTING

Product Backlog & Sprint Tasks

- Sprint-1 (Database Setup): Design the relational schema and import the 4,416 records into MySQL. (Story Points: 5, Priority: High)
- Sprint-2 (Data Connectivity & Cleansing): Connect Tableau via ODBC, rename technical columns, and build calculated fields. (Story Points: 5, Priority: High)
- Sprint-3 (Dashboard UI Development): Build interactive charts and cross-filtering views for target stakeholders. (Story Points: 8, Priority: High)
- Sprint-4 (Testing & Deployment): Perform cross-system data validation, review performance, and publish the dashboard. (Story Points: 2, Priority: Medium)

Sprint / Day	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Status / Release Date
Sprint-1	5	1 Day	03 Jul 2026	03 Jul 2026	Completed (Database Setup & Row Import)
Sprint-2	5	2 Days	04 Jul 2026	05 Jul 2026	Completed (Tableau Connection & Canvas)

Sprint / Day	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Status / Release Date
Sprint-3	8	1 Day	06 Jul 2026	06 Jul 2026	Completed (Dashboard Filters & Views)
Sprint-4	2	1 Day	07 Jul 2026	07 Jul 2026	Completed (Final Presentation & Submission)

Velocity Calculation

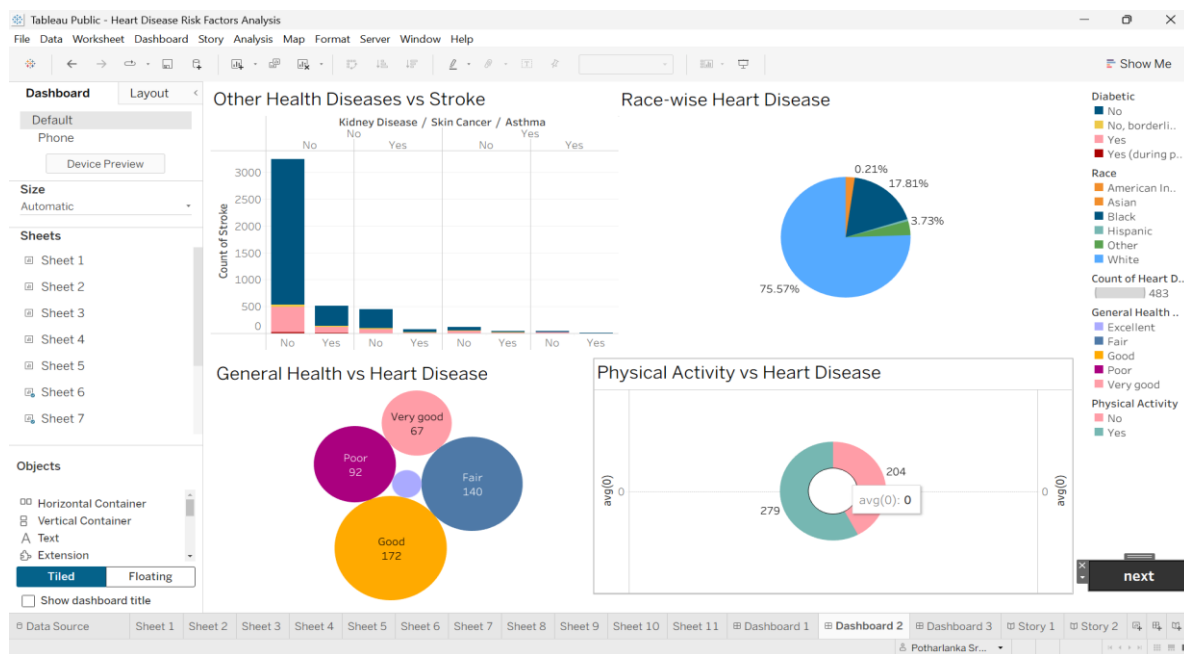
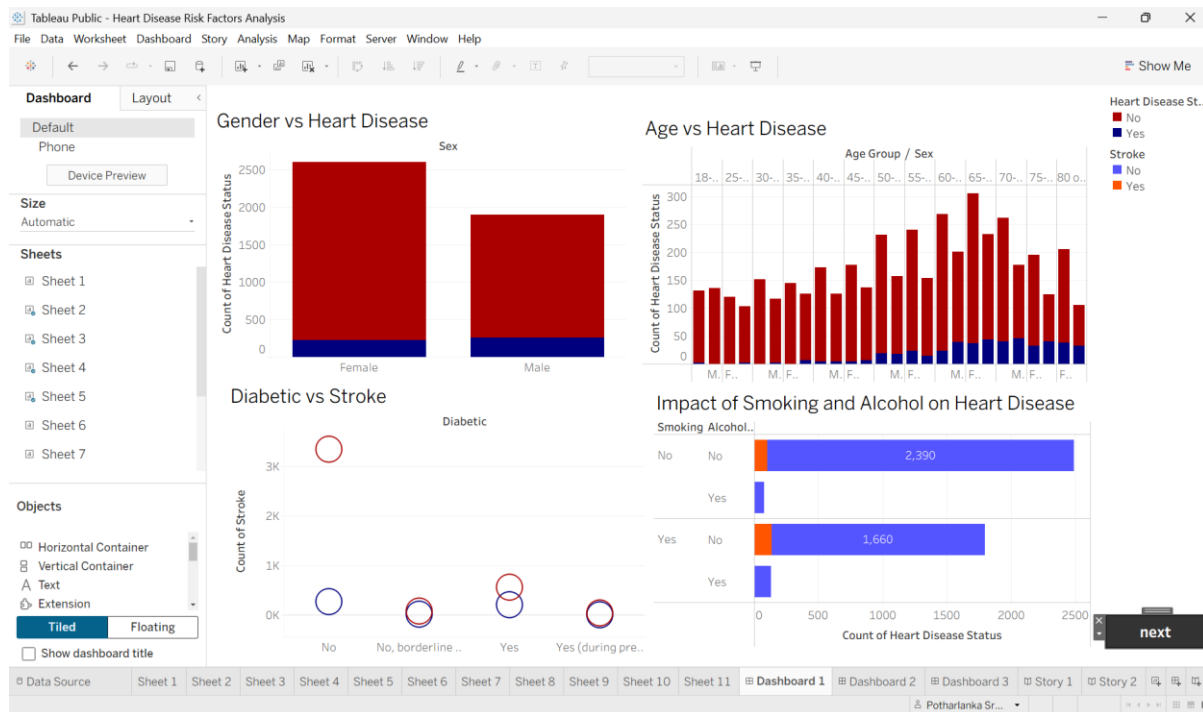
Using the standard template estimation formula, the team's average velocity (\$AV\$) per day for a standard 10-day sprint block covering 20 total story points is :

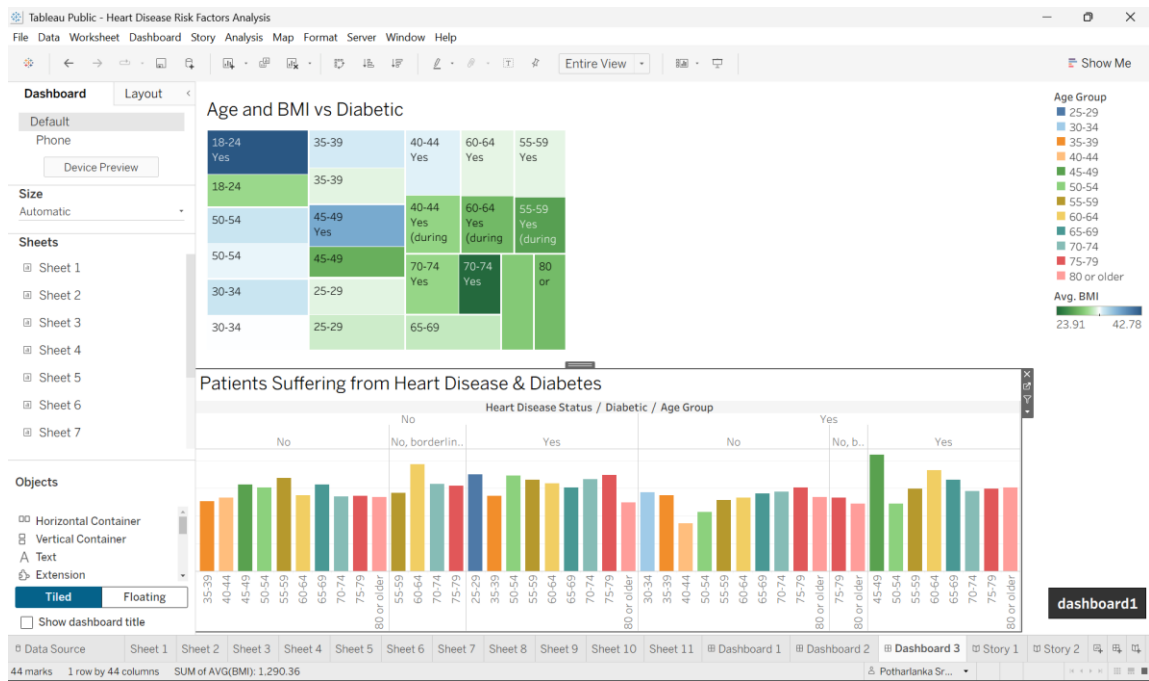
$$AV = \frac{\{\{\text{Team Velocity}\}\}}{\{\{\text{Sprint Duration}\}\}} = \frac{20}{10} = 2 \{ \text{story points per day} \}$$

6.1 Performance Testing

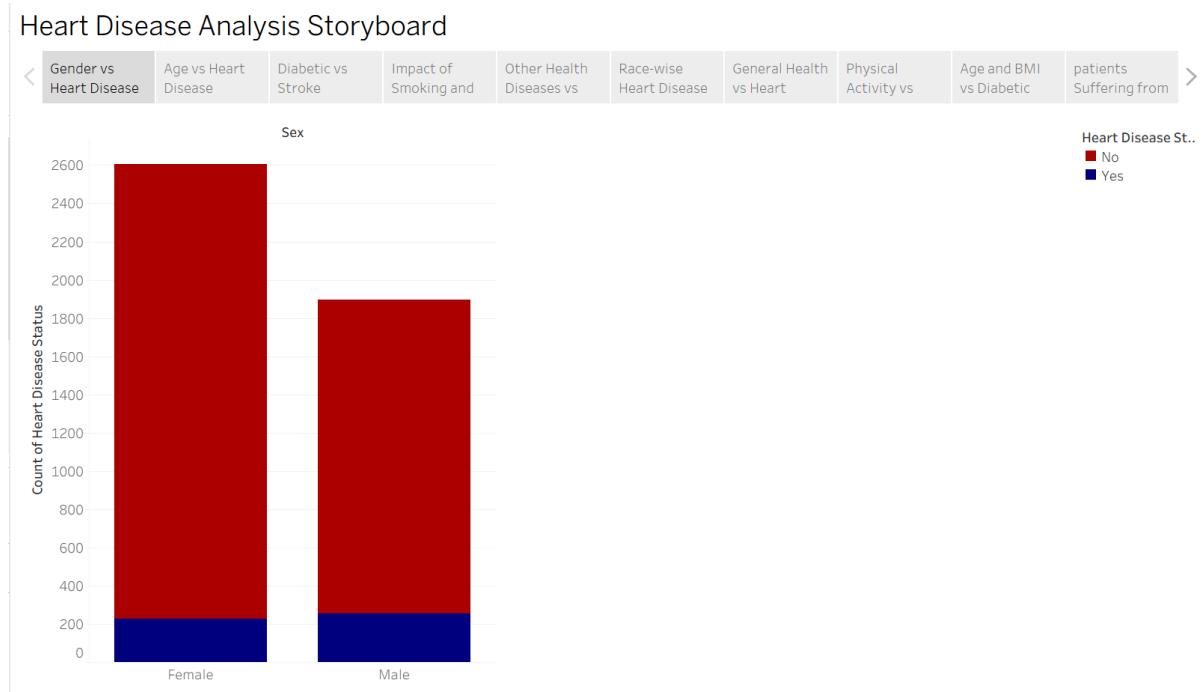
The screenshot displays the MySQL Workbench interface. On the left, the 'Navigator' pane shows the 'SCHEMAS' tree with 'data' expanded, and 'heart_new2' selected under 'Tables'. The main window shows the 'Table Details' for 'data.heart_new2'. The table is an InnoDB table with 18 columns and 4416 rows. The columns are: HeartDisease (text), BMI (double), Smoking (text), AlcoholDrinking (text), Stroke (text), and PhysicalHealth (int). The table size is estimated at 528.0 KiB. The data path is C:\ProgramData\MySQL\MySQL Server 8.0\Data\data\heart_new2.ibd, and it was created on 2023-06-05 at 15:19:42.

7.1 Dashboard Screenshots:

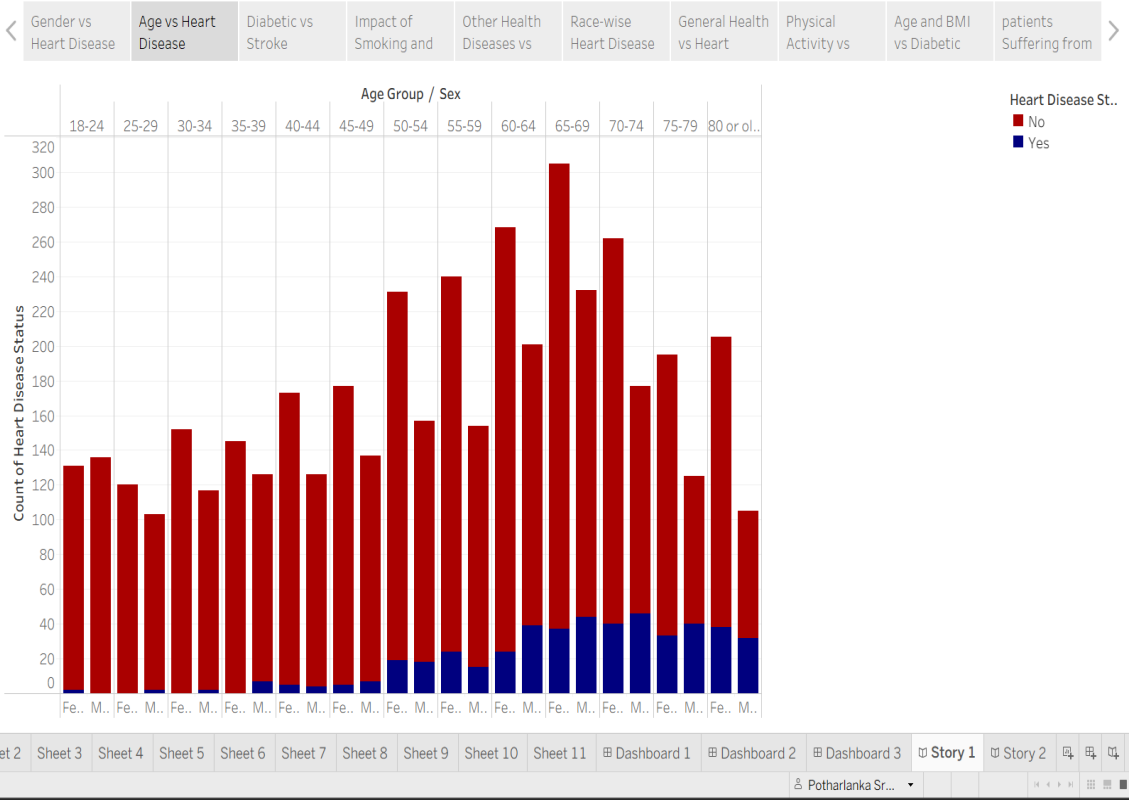




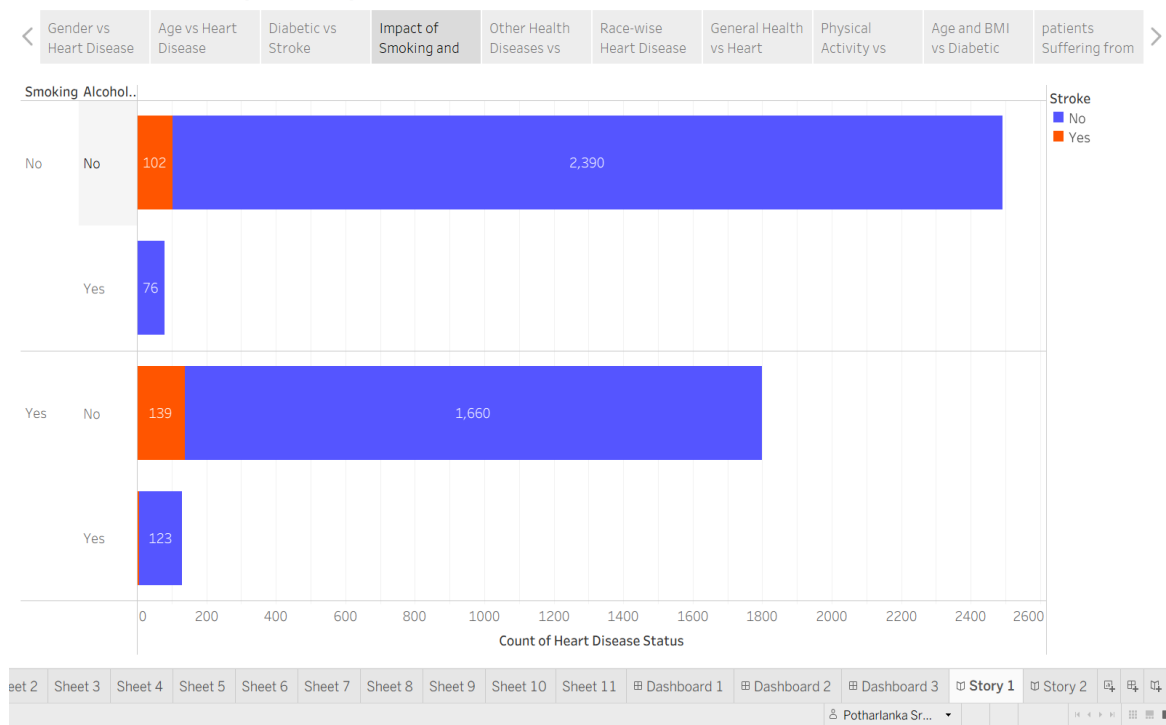
7.1 Story screenshots:



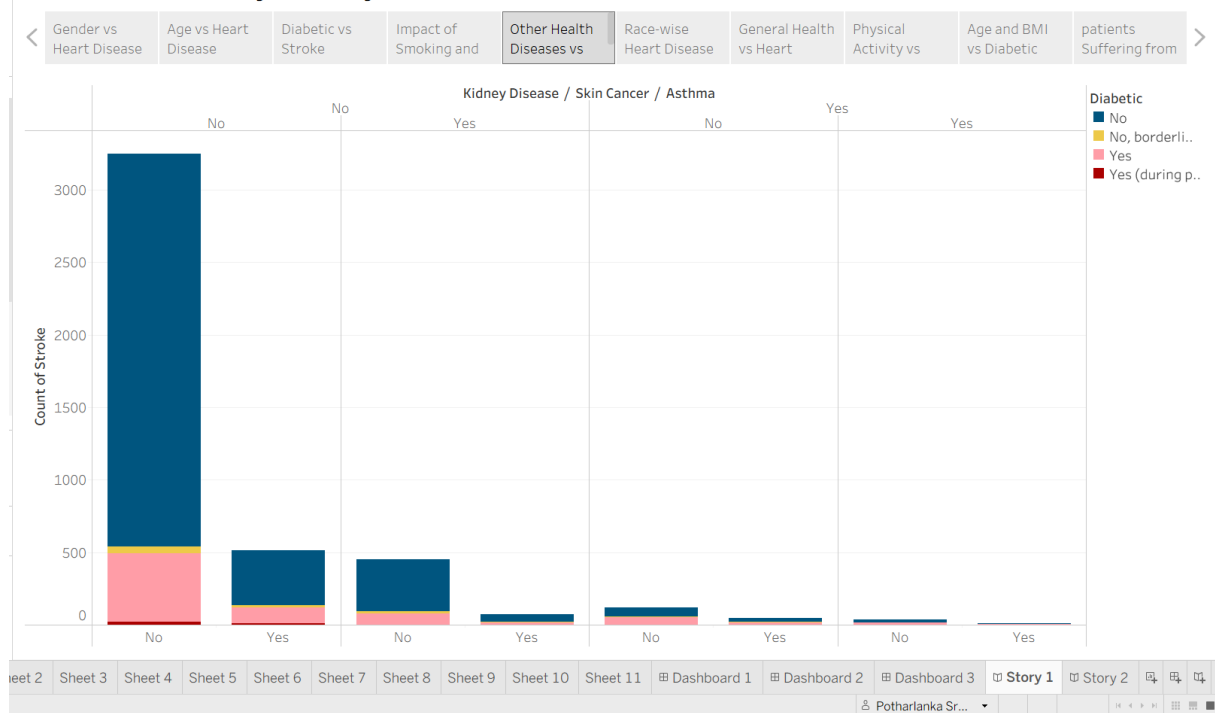
Heart Disease Analysis Storyboard



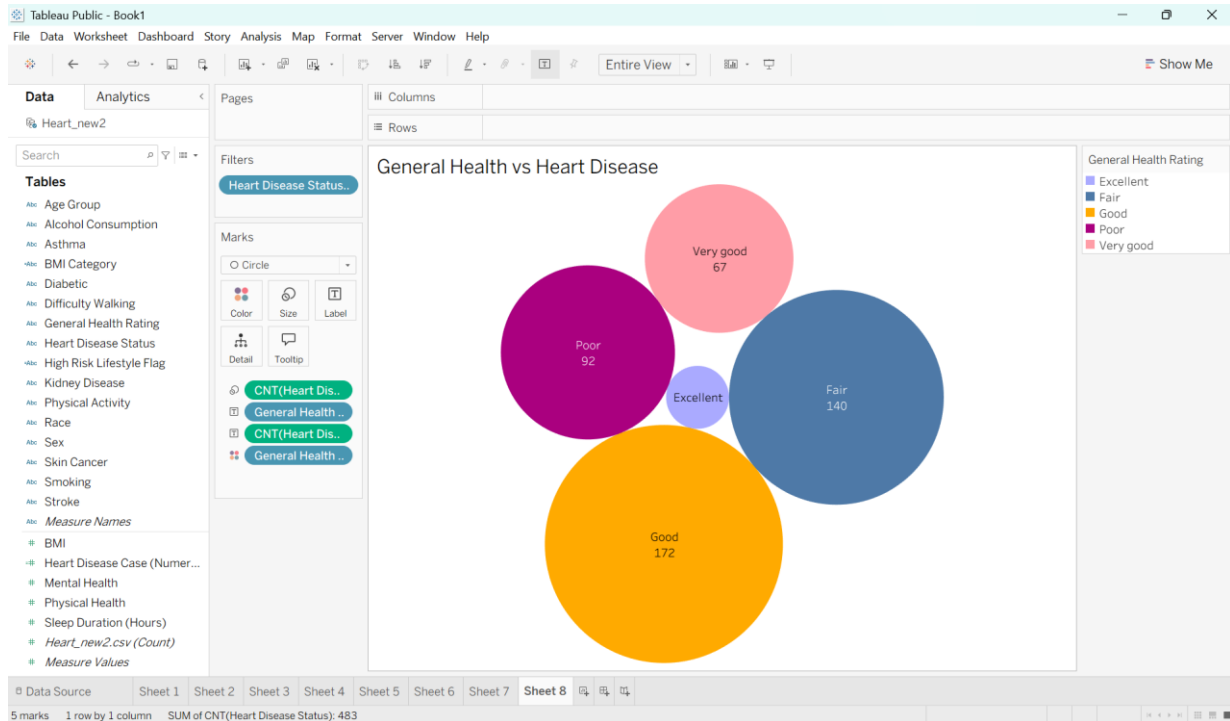
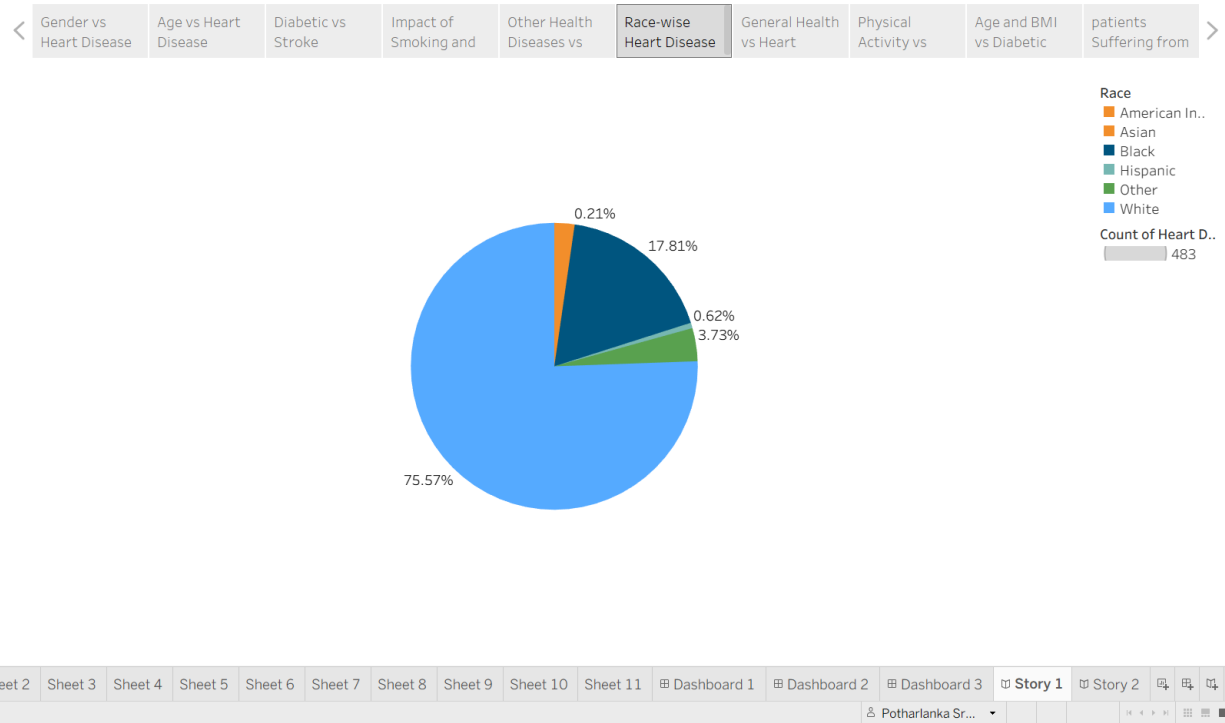
Heart Disease Analysis Storyboard



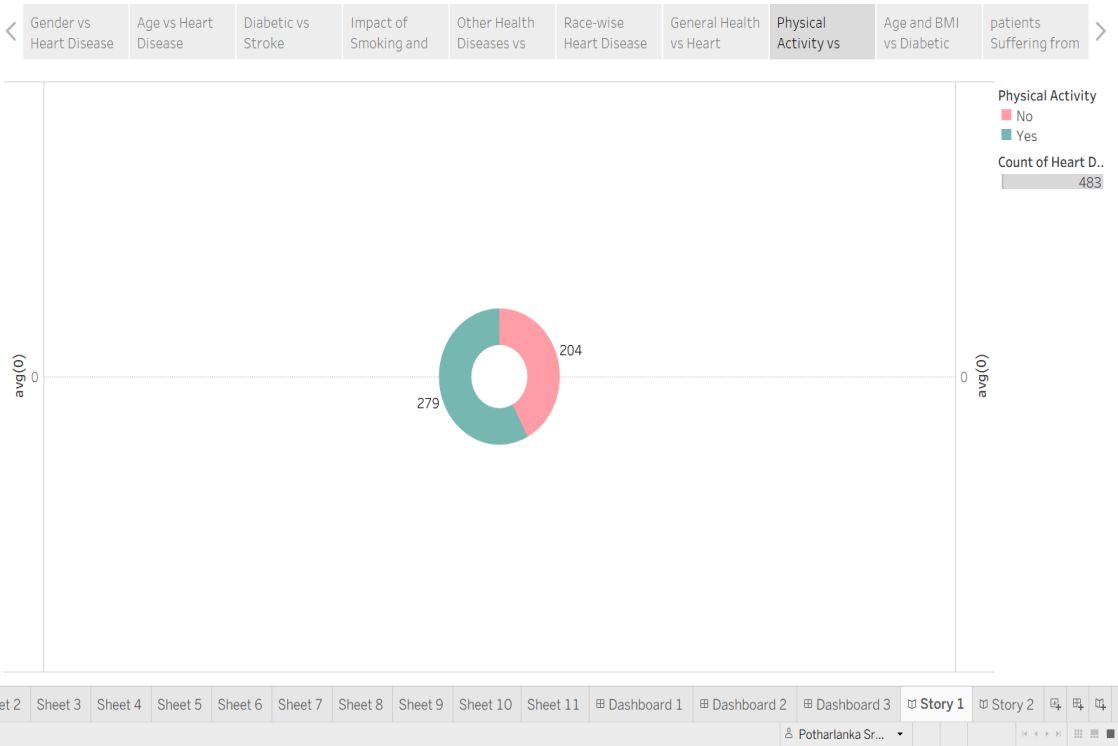
Heart Disease Analysis Storyboard



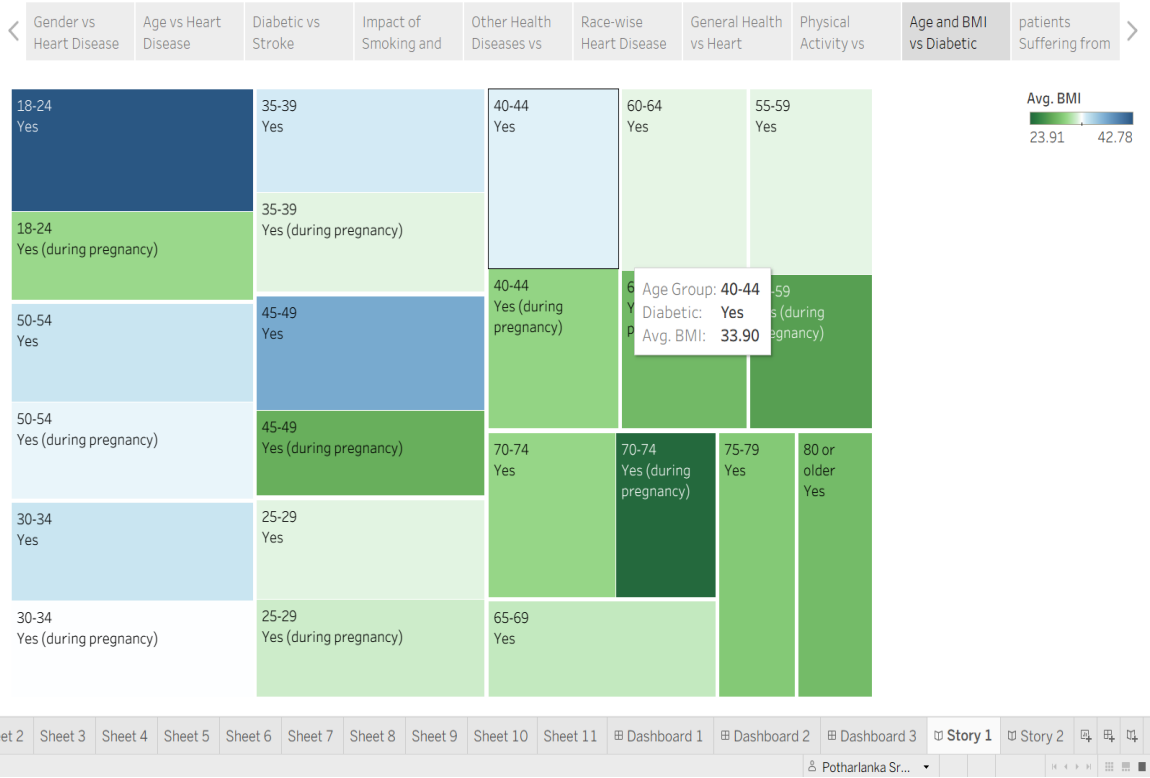
Heart Disease Analysis Storyboard



Heart Disease Analysis Storyboard



Heart Disease Analysis Storyboard



Heart Disease Analysis Storyboard



8. ADVANTAGES & DISADVANTAGES

• Advantages:

1. Replaces complex database queries with intuitive, 1-click cross-filtering charts.
2. Provides stakeholder-specific views designed for clinicians, public health officials, and patients.
3. Runs efficiently on local infrastructure using direct ODBC database connections.

• Disadvantages:

1. Relies on manual dataset updates rather than a real-time live patient telemetry feed.
2. Requires a local machine setup since it lacks an enterprise cloud application server .

9. CONCLUSION

The project successfully builds an analytics pipeline that translates raw clinical rows into clear visual insights. By bridging MySQL with Tableau, it provides a practical tool for data-driven healthcare analysis, helping stakeholders track risks and identify health trends across 4,416 patient records

10. FUTURE SCOPE

Future updates aim to connect the system to a live cloud data warehouse (such as AWS Snowflake or Google BigQuery) to support real-time data streaming. We also plan to integrate machine learning prediction models to calculate automated patient risk probabilities directly within the dashboard view.

11. APPENDIX

Source Code :

```
use collections;  
SELECT * FROM heart_new2;  
SELECT * FROM heart_new2 LIMIT 100;  
SELECT * FROM heart_new2 WHERE HeartDisease = 'Yes';
```

Dataset Link:

<https://drive.google.com/file/d/1VkiKCAFHejCZmS2OKW5KfMX1xzyPuGzz/view?usp=sharing>

GitHub link:

<https://github.com/sreejapotharlanka/Heart-Disease-Analysis>

Project Demo Link:

<https://drive.google.com/file/d/18itIvpfotPjnVNm3OqUDD2U68twI6zHy/view?usp=sharing>